Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 8 [2023 P2]

Test Date: 30th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2023-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. For positive integer 'n', define

$$f(n)=n+\frac{16+5n-3n^2}{4n+3n^2}+\frac{32+n-3n^2}{8n+3n^2}+\frac{48-3n-3n^2}{12n+3n^2}+.....+\frac{25n-7n^2}{7n^2}$$

Then the value of $\lim\limits_{n\to\infty} f(n)$ is equal to

A) $4+\frac{3}{4}\ln\frac{7}{3}$ **B)** $4-\frac{3}{4}\ln\frac{7}{3}$ **C)** $3+\frac{3}{4}\ln\frac{7}{3}$ **D)** $3-\frac{3}{4}\ln\frac{7}{3}$

2. For certain real numbers a, b and c the polynomial $g(x)=x^3+ax^2+x+10$ has three distinct roots, and each root of g(x) is also a root of the polynomial $f(x)=x^4+x^3+bx^2+100x+c$. The value of f(1) is ……

A) –7007 **B)** 7007 **C)** 4001 **D)** –4001

3. Number of solutions of equation

$\tan^3 x+\cot^3 x+8=(\tan x+\cot x+2)^3$ in $[0,4\pi]$ is …….

A) 4 **B)** 2 **C)** 8 **D)** 6

4. The area of the region $R_1\cap R_2\cap R_3$ where

$R_1=\left\{(x,y):\sqrt{x^2-y^2+8y}\geq 4\right\}$,

$R_2=\left\{(x,y):(x+y-4)(y-2x+2)\geq 0\right\}$,

$R_3=\left\{(x,y):2x^2-2y^2+3xy-24x+7y+22\geq 0\right\}$

A) 10 **B)** 6.5 **C)** 5 **D)** 4.5

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let $\hat{i}$, $\hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes. Let

$\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$,

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, b_2, b_3 \in \mathbb{R}$,

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, c_1, c_2, c_3 \in \mathbb{R}$

Be three vectors such that $b_2b_3 > 0$, $\vec{a}.\vec{b} = 0$ and $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$

Then, which of the following is/are true?

A) $\vec{a}.\vec{c} = 0$ **B)** $\vec{b}.\vec{c} = 0$ **C)** $|\vec{b}| < \sqrt{10}$ **D)** $|\vec{c}| \le \sqrt{11}$

6. Let $a_1, a_2, a_3, \ldots\ldots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3\ldots\ldots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \ge 1$. Then, which of the following is/are True?

A) $T_{20} = 1504$ **B)** $\sum_{k=1}^{20} T_k = 10510$ **C)** $T_{30} = 3454$ **D)** $\sum_{k=1}^{30} T_k = 35610$

7. $P_1 : x + 3y + 5z = 2$, $P_2 : x - 2y + z = 5$ are two planes. From a point P(-4, 7, 11) perpendiculars PQ and PR are drawn to planes P_1 and P_2 respectively. Then

A) The direction ratios of line represented by planes P_1 and P_2 may be equal to (13, 4, 5)

B) Area of triangle PQR = $2\sqrt{210}$ sq. unit

C) Mid point of QR = (-2, 3, 13)

D) Equation of plane PQR is 13x + 4y – 5z + 79 = 0

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha+\beta) = \frac{1}{3}$ and $\cos(\alpha-\beta) = \frac{2}{3}$. If $\lambda = \left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\cos\alpha}{\sin\beta} + \frac{\sin\beta}{\cos\alpha}\right)^2$, then $\left|\frac{9\lambda}{4}\right|$ equals

9. Consider the hyperbola $\frac{x^2}{100}-\frac{y^2}{64}=1$ with foci at S and S_1, where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1=\alpha$, with $\alpha<\frac{\pi}{2}$. The straight line passing through the points S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line S_1P at P_1. Let δ be the distance of P from the straight lines SP_1 and $\beta=S_1P$. If $t=\beta\delta\sin\frac{\alpha}{2}$, then value of $\frac{t}{8}$ equals.

10. Let z_1 and z_2 be complex number such that $z_1^2-4z_2=16+20i$ and the roots α and β of $x^2+z_1x+z_2+m=0$ for some complex number 'm' satisfies $|\alpha-\beta|=2\sqrt{7}$; if the sum of maximum and minimum value of |m| is equal to λ, then 2λ is …….

11. There are 20 observations $x_1, x_2,....., x_{20}$. If mean of $(x_i+i)\forall\, i=1, 2, 3, ...,20$ is 15.5 and $\sum_{i=1}^{20}(x_i-5)^2=100$. If variance of $(x_i-2)\ \forall i=1, 2, 3, ...,20$ is λ, then λ is equal to

12. A fair coin is flipped 'n' times let E be the event "a head is obtained on first flip" and F_k be the event "exactly 'k' heads are obtained", then the value of n/k for which events E and F_k are independent is :

13. Let a, b, c be three natural numbers such that $a^3-b^3-c^3=3abc$ and $a^2=2(b+c)$ and 'p' be an integer for which both roots of the quadratic equation $x^2+a(p-(a+b))x+9=0$ lies in the interval $(-(a+b+3c), b)$, if a, g_1, g_2,....., g_{20}, p are in geometrical progression then the value of g_4g_{17} equals …….

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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PARAGRAPH-I

Let a, b and c are real numbers such that the system of linear equations

$x + y + z = a$

$x + 5y – z = b$

$x – y + 2z = c + 2$ is consistent.

Let |M| represent the determinant of the matrix $M=\begin{bmatrix}a & -2 & b\\ 2 & 1 & c\\ 1 & 0 & 3\end{bmatrix}$. Let 'P' be the plane containing all those (a, b, c) for which above system of linear equations is consistent and 'D' be the square of distance of the point (1, 1, 1) from the plane 'P'.

14. The value of $\frac{\sqrt{7}}{5\sqrt{2}}$ D is ….

15. The absolute value of |M| is …..

PARAGRAPH-II

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to \mathbb{R}$, $i = 1, 2$ and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to \mathbb{R}$ be functions such that

$g_1(x) = 1$, $g_2(x) = |4x - \pi|$ and $f(x) = \sin^2 x$, for all $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$

Define $S_i = \int_{\frac{\pi}{16}}^{\frac{7\pi}{16}} f(x).\, g_i(x)\, dx$, $i = 1, 2$

16. The value of $\frac{16S_1}{\pi}$ is…..

17. The value of $\frac{64S_2}{\pi^2}$ is ……

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

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18. A car is fitted with a convex mirror of focal length 20cm. a second car 2m broad and 1.6m high is 6m away from the first car. The position of the second car as seen in the mirror of the first car is

A) 19.35cm **B)** 17.45cm **C)** 21.48cm **D)** 5.49cm

19. Two charges $12\mu C$ and $-6\mu C$ are separated by a distance of 20cm. The net electric potential is zero on the line joining them.

A) at a distance of 6.7cm from $-6\ \mu C$ in between the charges

B) at a distance of 0.2m from $-6\ \mu C$ outside the charges

C) both of the above

D) none of the above

20. A proton, a deuteron and an α -particle are accelerated by the same potential difference. Their velocities gained will be in the ratio of

A) 1: $\sqrt{2}$:1 **B)** 1:1: $\sqrt{2}$ **C)** $\sqrt{2}$ **:**1:1 **D)** 1:1:1

21. A body of mass 0.4kg is whirled in a vertical circle making 2 rev/sec. If the radius of the circle is 1.2m, then tension in the string when the body is at top of the circle is

A) 71.88N **B)** 89.86N **C)** 109.86N **D)** 115.86N

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

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22. Two pulses travelling on same string are described by equations.

$$Y_1 = \frac{5}{(3x-4t)^2+2} \text{ and } Y_2 = \frac{-5}{(3x+4t-6)^2+2}$$

Mark out **CORRECT** statements

A) At $t = \frac{4}{3}s$ the two waves cancel each other everywhere

B) At $t = \frac{3}{4}s$ the two waves cancel each other everywhere

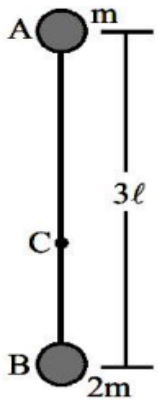
C) At $x = 1$ the two waves cancel each other at all times

D) At all points two waves cancel each other

23. The length of a strip measured with a meter rod is 10.0 cm. Its width measured with a vernier callipers is 1.00 cm. The least count of the meter rod is 0.1 cm and that of vernier callipers is 0.01 cm. What can be the error in its area?

A) $+0.01\text{cm}^2$ **B)** $-0.1cm^2$ **C)** $+0.2\,cm^2$ **D)** $-0.02\,cm^2$

24. A light rigid rod AB of length 3l has a point mass 'm' at end A and a point mass 2m at end B is kept on a smooth horizontal surface. Point C is the center of mass of the system. Initially the system is at rest. The mass 2m is suddenly given a velocity v_0 towards right. Take Z axis to be perpendicular to the plane of the paper.



A) The minimum moment of inertia (about Z-axis), I_{zz} of the system is 5ml^2.

B) The magnitude of tension in the rod in subsequent motion is $\frac{2mv_0^2}{9l}$

C) The ratio of moment of inertia about Z-axis at points A and B, $\frac{I_{ZZ}^{A}}{I_{ZZ}^{B}} = 2$

D) Point C remains stationary during subsequent motion

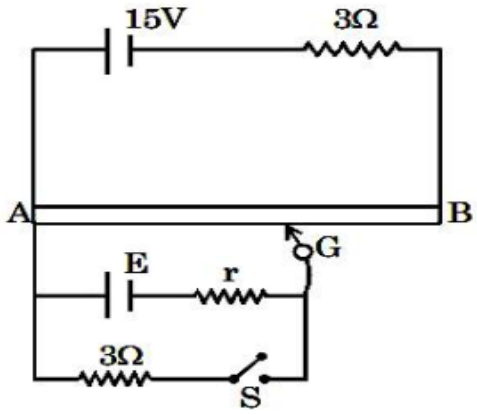
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
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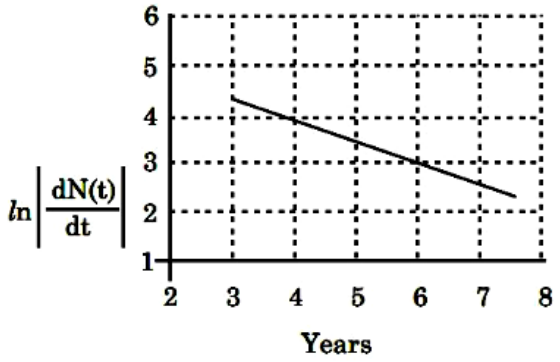
25. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage V by angle 30^0. When the same resistance is connected in series with element B, current leads by 60^0. When R, A, B are connected in series, the current now leads voltage by θ which is equal to $\tan^{-1}\left(\frac{K}{\sqrt{3}}\right)$, then the value of K is (assume same AC source is used in all cases)

26. A wire AB (of length 1m, area of cross section πm^2) is used in potentiometer experiment to calculate emf and internal resistance (r) of battery. The emf and internal resistance of driving battery are 15 V and 3Ω respectively. The resistivity of wire AB varies as $\rho = \rho_0 x$, where 'x' is distance from A in meters and $\rho_0 = 24\pi\Omega$. The distance of null point from A is obtained at $\sqrt{\frac{2}{3}}$ m when switch 'S' is open. Find the value of E.

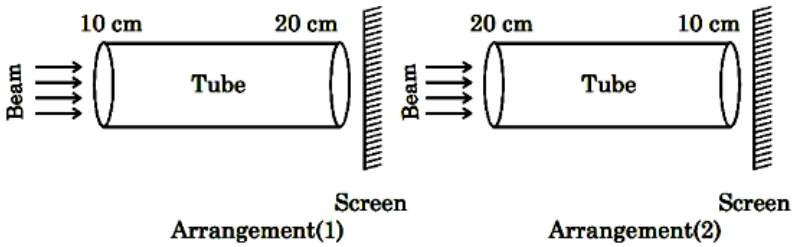


27. To determine the half life of a radioactive element, a student plots a graph of $\ln\left|\frac{dN(t)}{dt}\right|$ versus 't'. Hence $\frac{dN(t)}{dt}$ is the rate of radioactive decay at time 't'. If the number of

radioactive nuclei of this element decreases by a factor of 'p' after 4.16 years, the value of 'p' is:



28. A tube of length 30 cm has its inner lateral surface blackened. Two lenses of focal lengths 10cm and 20 cm are fixed at the ends of the tube. For same beam, intensities of spots on screen are I_1 and I_2 for arrangement (1) and arrangement (2). If the ratio of $I_2 : I_1$ is of the form N^2, find N.



29. A solid copper sphere (e = 1, density ρ and specific heat capacity c) of radius 'r' at an initial temperature 200 K is suspended inside a chamber whose walls are at almost 0K. The time required (in μs) for the temperature of the sphere to drop to 100 K is $\frac{n}{72}\frac{r\rho c}{\sigma}$. Find the value of 'n'. (All symbols have their usual meaning)

30. A glass surface is coated by an oil film of uniform thickness 1.00×10^{-4} cm. The index of refraction of the oil is 1.25 and that of the glass is 1.50. Some of the wavelengths in visible region (400 nm – 490 nm) are completely transmitted by the oil film under normal incidence. One of the wavelength transmitted completely in visible region is $\frac{n\times10^{-6}}{11}$m. Find the value of 'n'.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
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PARAGRAPH-I

To find the expansion of length (ΔL) of a rail track, one can use the formula :

$\Delta L = L_{\alpha}\Delta T$

Here

L : original length

α : coefficient of linear expansion

ΔT : Rise in temperature

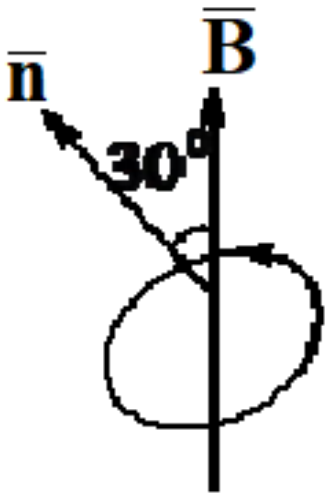
Consider a rail track of length 20 m at $10^0 C$ having coefficient of linear expansion $\alpha = 12\times10^{-6}/{}^0C$ When the wheels of train cross the gap between the two such rail tracks, we hear rhythmic sound (thak thak) and distance between the two consecutive wheels along the rail track is also 20m. Suppose we find that wheels are making 75 clicks (rhythmic sound) in a minute

31. Estimate the speed of train (in km/hr)

32. Find the magnitude of change in length of rail track when temperature rises to $60^0 C$ (in mm)

PARAGRAPH-II

An electron in the ground state of hydrogen atom is revolving in anticlockwise direction in a circular orbit of radius R (shown in figure)



The orbital magnetic moment of the electron is *M*. The atom is placed in a uniform magnetic induction $\vec{B}$-such that the plane normal of the electron orbit makes an angle of

30^0 with the magnetic induction. The torque experienced by the orbiting electron is $\vec{\tau}$. Based on the above facts, answer the following questions. **"Bohr's model is applicable"**

33. The magnitude of $\vec{M}$ is $\frac{eh}{k\pi m}$, (upwards and perpendicular to the plane of the orbit).

Find the value of 'k'

34. The magnitude of $\vec{\tau}$ is $\frac{eBh}{k\pi m}$. Find the value of 'k' is

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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35. A sol is prepared by adding lower amount of silver nitrate solution to dilute potassium iodide solution, then

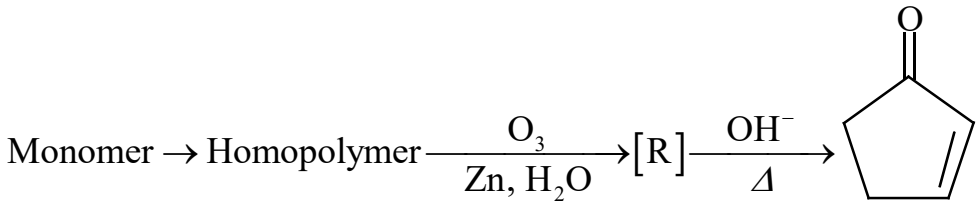
A) Sol particles move towards anode during electro-osmosis.

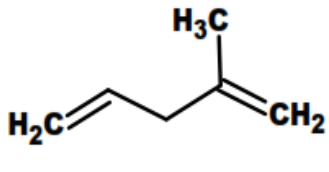
B) Dispersion medium moves towards cathode during electrophoresis.

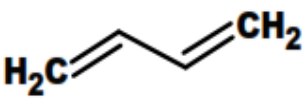
C) Silver iodide precipitate adsorbs iodide ions to form sol.

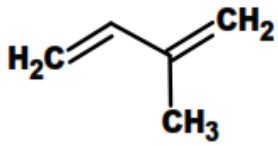
D) Silver iodide precipitate absorbs iodide ions to form sol.

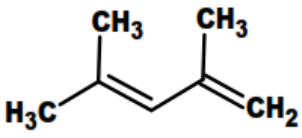
36. Identify the monomer in this sequence



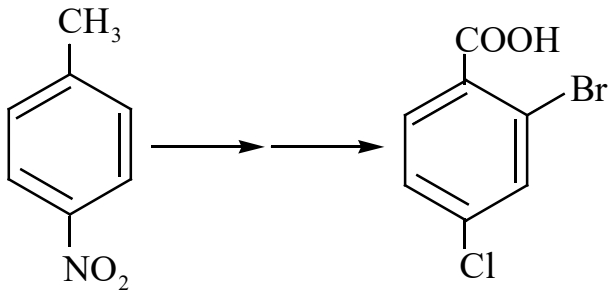
A)


B)


C)


D)


37.



In the above conversion the correct sequence of reagents to be added is

A) (i) Fe/H^+, (ii) HONO, (iii) CuCl, (iv) $KMnO_4$, (v) Br_2

B) (i) $KMnO_4$, (ii) Br_2/Fe, (iii) Fe/H^+, (iv) Cl_2

C) (i) Br_2/Fe, (ii) Fe/H^+, (iii) HONO, (iv) CuCl, (v) $KMnO_4$

D) (i) Br_2/Fe, (ii) Fe/H^+, (iii) $KMnO_4$, (iv) Cl_2

38. Identify the incorrect statement:

A) In ClF_3 molecule, angle of FClF is 90^0

B) In XeO_6^{4-} all bond length are equal

C) In NF_3 and NH_3 bond length N – F > N – H

D) Bond angle in NH_3 is more than NF_3

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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39. In an evacuated vessel of capacity 112 L, 4 moles of Ar(g) and 5 moles of $PCl_5(g)$ were introduced and the temperature is maintained at 273^0C . At equilibrium, the total pressure of the mixture was found to be 4.8 atm.

A) The degree of dissociation of $PCl_5(g)$ into $PCl_3(g)$ and $Cl_2(g)$ is 0.6

B) K_P for the reaction : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 2.7 atm

C) At equilibrium, the total moles of gases are 12

D) On removing Ar(g) from the equilibrium mixture at constant pressure and temperature, the extent of dissociation of PCl_5 will increase

40. In which of the following reaction/s correct rate of reaction is/are?

A) $\underset{r_1}{\leftarrow (CD_3)_3C-Cl} \; (C_2H_5OH + KOH)/\Delta \; \underset{r_2}{(CH_3)_3C-Cl \rightarrow} \; (r_2 > r_1)$

B) $\underset{r_1}{\leftarrow C_6D_6/\Delta} \; \text{conc. } HNO_3 + \text{conc. } H_2SO_4 \; \underset{r_2}{C_6H_6/\Delta \rightarrow} \; (r_2 > r_1)$

C) $\underset{r_1}{\leftarrow C_6D_6/\Delta} \; \text{conc. } H_2SO_4 \; \underset{r_2}{C_6H_6/\Delta \rightarrow} \; (r_2 > r_1)$

D) $\underset{r_1}{\leftarrow CD_4/\Delta} \; Cl_2/hv \; \underset{r_2}{CH_4 \rightarrow} \; (r_2 > r_1)$

41. Which of the following is correct order/s for indicated bond length …

A) d_{N-N} in $N_2H_4 > d_{N-N}$ in N_2F_4 **B)** d_{N-N} in $NH_2-NH_2 > d_{N-N}$ in $\overset{\oplus}{N}H_3-\overset{\oplus}{N}H_3$

C) d_{Si-Cl} in $SiClF_3 > d_{Si-Cl}$ in $SiClH_3$ **D)** d_{C-C} in $C_2H_6 > d_{C-C}$ in C_2F_6

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

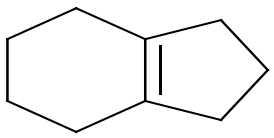
42. Mole fraction of N_2 gas in air is 0.8. Henry law constant for N_2 in water is 2.22×10^4 atm. The moles of N_2 gas present in 1L of solution which is in equilibrium with air at 10 atm is $a\times10^{-2}$ find 'a' ?

43. During the discharge of a lead storage battery, the density of H_2SO_4 fell from 1.225 gm/ml to 0.98 gm/ml. H_2SO_4 of density 1.225 gm/ml is 40% by weight and that of density

0.98 gm/ml is 20% by weight. The number of ampere hours for which the battery is used containing 1 L of the acid is 16.08 x. Calculate the value of x. (volume remains constant during the process)

44. Total number of cyclic isomers (excluding stereo isomers) possible for C_4H_8O which do not liberate H_2 gas with Na metal is……

45. The Degree of Unsaturation (DU) of major product (X) in the following reaction is



$$\xrightarrow[\text{(iii) } BF_3.Et_2O]{\text{(i) NBS / } H_2O \\ \text{(ii) } K_2CO_3\text{(aq)}} (X)$$

(Major)

46. Find the total number of compounds which on heating undergo redox reactions.

$PbCl_4$, $Mg(NO_3)_2$, HgC_2O_4, Ag_2CO_3, $Pb(CN)_4$, $Al(OH)_3$, $Cu(CN)_2$

47. Find total number of complex in which metal – carbon linkage(s) is/are present.

$[Fe(EDTA)]^-$, $K[PtCl_3(\pi-C_2H_4)]$, $[Cr(acac)_3]$, $[Fe(CO)_5]$, $[Co(C_2H_5O)_6]^{3-}$

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Amount of heat evolved during complete combustion of liquid benzene can be calculated from the following data.

I) 18 gm of graphite on complete combustion evolves 591 kJ heat

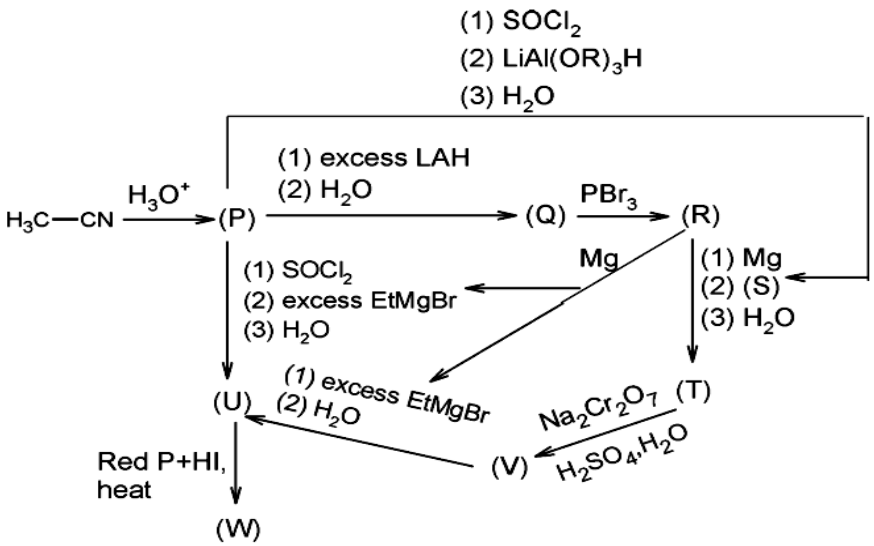
II) $\Delta H_f^0(H_2O, l) = -286$ kJ / mol

III) The heat of formation of liquid benzene is 50 kJ /mole

48. Magnitude of Heat of formation of $CO_2(g)$ from following data (in kJ/mole) is …….

49. Magnitude of heat evolved (in kJ) from combustion of 78 gm benzene (*l*)

PARAGRAPH-II



50. Total number of compounds from (P) To (U) which can give yellow ppt when react with NaOI?

51. Total number of monochloro products possible for product (W) is. (including stereo)